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Accuracy and reproducibility of latent print decisions on comparisons from searches of an automated fingerprint identification system

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ABSTRACT

This study was conducted to evaluate the accuracy and reproducibility of decisions made by practicing latent print examiners (LPEs) when comparing latent fingerprints to exemplars that were acquired by searches of the FBI Next Generation Identification system (NGI). This study builds on the 2009 FBI-Noblis latent print examiner black box study (1, 2), incorporating lessons learned from conducting multiple additional forensic examiner studies. Analyses are based on 14,224 responses from 156 LPEs. Each participant was assigned 100 latent-exemplar image pairs (IPs, 80 nonmated and 20 mated) out of a total of 300 IPs. On mated comparisons, 62.6 % of responses were IDs (true positives), 4.2 % were erroneous exclusions (false negatives), 17.5 % were inconclusive, and 15.8 % were no value. On nonmated comparisons, 0.2 % of responses were erroneous IDs (false positives), 69.8 % were exclusions (true negatives), 12.9 % were inconclusive, and 17.2 % were no value. One participant made the majority of the erroneous IDs in the study, which underscores how decision rates can be highly sensitive to the individual participants in a study. No erroneous IDs were reproduced by different LPEs, but 15 % of erroneous exclusions were reproduced. There is a potential concern that NGI's size and ability to yield much more similar nonmates (as compared to IAFIS used in (1, 2)) poses an increased risk of false IDs; however, we do not find evidence of an increase in the observed false ID rate. These results suggest that risk mitigation strategies for NGI searches may be working for those agencies that have implemented such strategies.

1. Introduction

Latent prints ("latents") have been extensively used in legal proceedings for over a century and remain one of the most common types of forensic evidence examinations requested of forensic laboratories [1,2]. Despite the longstanding historical precedent and general acceptance of latents as evidence, reports published within the last several decades have called for robust assessments of the scientific validity of the decisions made in forensic pattern-based disciplines, including fingerprints [3–6]. The scientific validity of latent fingerprint evidence is based on the accuracy and reliability of decisions made by latent print examiners (LPEs), who compare questioned impressions or latents (e.g., crime scene impressions) to exemplar fingerprints collected from suspects and/or other known individuals, with the goal of rendering an opinion about source attribution.

The 2016 President's Council of Advisors on Science and Technology (PCAST) report stated that multiple independent black box studies (involving a sufficiently large number of examiners, representative samples, and relevant study populations) are necessary to establish scientific validity for subjective feature comparison methods: "the only way to establish the scientific validity and degree of reliability of a subjective forensic feature-comparison method—that is, one involving significant human judgement—is to test it empirically by seeing how often examiners actually get the right answer" [5]. With respect to fingerprint evidence, the report held up two studies as examples that met the criteria for robustly evaluating the accuracy and reliability of LPE decisions: the original FBI-Noblis study [7,8] (conducted in 2009, hence hereafter LatentBB09) and the Miami-Dade study [9].

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Here, we present the results of the LPE Black Box Study 2022 (hereafter LatentBB22), which was conducted to evaluate the accuracy and reproducibility of LPE decisions when comparing latents to exemplar fingerprints that were acquired by searches of the FBI's Next Generation Identification system (NGI). Of particular interest is the extent to which the results measured here compare to those from the original LatentBB09 study, which was conducted prior to the advent of NGI. The study design incorporates lessons learned since the LatentBB09 study regarding study design, implementation, and data analyses, based on the study team's experience conducting a variety of forensic examiner performance studies in the intervening years [10–16].

1.1. Background

A large portion of latent print casework involves searches of automated fingerprint identification systems (AFIS). When using an AFIS in latent print casework, an LPE electronically submits a latent (as an image alone or with marked minutiae) to an AFIS hosted by a local, state, or federal government agency. An AFIS uses complex proprietary algorithms to assess the submitted latent against all exemplar fingerprints in the database and returns a response with a list of the most similar (candidate) exemplar images; the examiner then compares the latent to the candidate exemplars to determine if any of the exemplars can be identified with the latent. The examiner may compare the entire candidate list or may limit which candidates are compared based on agency policy (and in some scenarios, examiner discretion). The individual examiner's agency—not the agency hosting the AFIS—is responsible for finalizing any decisions made with respect to a latent search, including verification, quality assurance, and reporting. Note that latent AFIS acts as a tool for finding the most similar exemplars in the database: human examiners and their agencies retain full responsibility for making identifications.[1] NGI [17] is the largest AFIS in the U.S. and is accessible to U.S. law enforcement agencies (local, state, territorial, tribal, or Federal). NGI is used frequently by LPEs: in the fiscal year ending Sept 2022, NGI received 317,228 latent searches [18].

Selection of fingerprint pairs in LatentBB09 was largely based on the FBI's Integrated Automated Fingerprint Identification System (IAFIS), whereas selection of fingerprint pairs in LatentBB22 was explicitly based on the subsequent FBI NGI system. The accuracy of automated latent fingerprint algorithms improved significantly between IAFIS and NGI: searches in NGI are now much more likely to return mates as candidates, and poor-quality latents are also more likely to be successfully searched [19]. In addition, the FBI's fingerprint repository has significantly increased the number of records that it contains: when the LatentBB09 nonmated image pairs were selected in 2009, the IAFIS database contained approximately 58 million subjects (580 million distinct fingers); when the LatentBB22 image pairs were selected in 2021, the NGI database contained approximately 128 million subjects (1.28 billion distinct fingers). The concern that searches of large AFIS databases may return highly similar nonmated candidates has been discussed in various sources [2, 20–22]; such a search was found to have contributed to the Madrid/Mayfield latent print misidentification [23]. Note also that with IAFIS, one rolled impression from each finger was included in the database; with NGI, multiple rolled and plain impressions from each finger are included in the database search. As a result of these changes, when searches are run using latents from subjects who are not in the NGI repository, we can expect the highest-ranking nonmated candidates returned by NGI to be much more similar to the latent searched than those returned by IAFIS. This raises the potential concern that LPEs using NGI are being asked to conduct more challenging comparisons, which in turn may affect the accuracy or reproducibility of their decisions.

[1] Note that latent AFIS workflow is very different from AFIS workflow for exemplars, such as for arrests or background checks.

This study follows the "black box" approach used by the authors in previous studies assessing decisions of forensic examiners in latent print comparison [7], bloodstain pattern analysis [16], handwriting comparison [15], and footwear comparison [14]. A forensic black-box study is an empirical test of forensic examiners rendering decisions on a series of independent comparisons of known-source samples selected to be broadly representative of operational casework, to assess the accuracy and reliability of their decisions without regard to how they make those decisions.

2. Materials and methods

This section summarizes the study design, participants, and the fingerprint data; see Appendix B for additional details.

2.1. Study overview

Each participant was assigned 100 latent-exemplar image pairs (IPs), each of which included one latent fingerprint image and one exemplar image (rolled or plain fingerprint from a ten-print set) for comparison: 20 of the IPs assigned to each participant were mated (known to be from the same subject and finger), and 80 were nonmated (known to be from different subjects). Participants were not provided any information indicating the proportions of mated vs. nonmated IPs. Each IP followed the workflow shown in Fig. 1. The test was administered using custom web-based software that presented examiners with digital images and collected all responses. The response for each IP was submitted prior to the next IP being shown; participants could not skip comparisons nor select the order to conduct comparisons. Like the original LatentBB09 study, this study focused on the key decision points during examination by individual LPEs. Note that this study does not include additional steps that may occur during operational latent examinations that may be implemented to reduce potential errors, such as examination of original evidence or paper fingerprint cards, review of multiple exemplars from a subject, and consultation with other examiners—as well as additional steps conducted after the LPE's comparison, including verification by other examiners, technical review, and quality assurance review.

The conclusion scales used by LPEs have changed over the past few decades. Prior to the 1990s, conclusions were generally Boolean: "identification" vs. "non-identification." By the 1990s, this was expanded (in most agencies) to explicitly differentiate "exclusion" from "inconclusive." There has been movement over the past few years to expand the conclusion scale to provide greater subtlety in responses and more precisely reflect the examiner's decisions: a 5-level scale was proposed by OSAC, which is currently a draft Academy Standards Board (ASB) standard [24]. The approach shown in Fig. 1 allows our analyses to treat the responses as a traditional 3-level conclusion scale (*Excl, Inc, ID*), as a 5-level OSAC/ASB conclusion scale (*Excl, Inc-Dis, Inc-No, Inc-Sim, ID*), or as a 7-level conclusion scale (*Excl-NB, Excl-B, Inc-Dis, Inc-No, Inc-Sim, ID-B, ID-NB*). The decision categories were adapted from the 5-level ASB 2021 draft standard conclusion scale [24] (which includes the three subcategories of inconclusive but does not differentiate between borderline and non-borderline decisions). See Appendix C for details of the questions asked.

In our analyses, we often have reason to group decisions using the following abbreviations: *NoValue* refers collectively to *NV-L* and *NV-E*; *VCMP* (value for comparison) refers collectively to *OfValue* and *LimValue*; *conclusive* decisions refer collectively to *ID* and *Excl*; *indeterminate* decisions (abbreviated *IncNV*) refer collectively to *Inc* and *NoValue*. Note that *NoValue* can be seen as a subset of an inconclusive decision in that it functions as a preemptive inconclusive decision, because regardless of the exemplar, an examiner making a *NoValue* decision is predicting that they would not be able to make a conclusive decision.

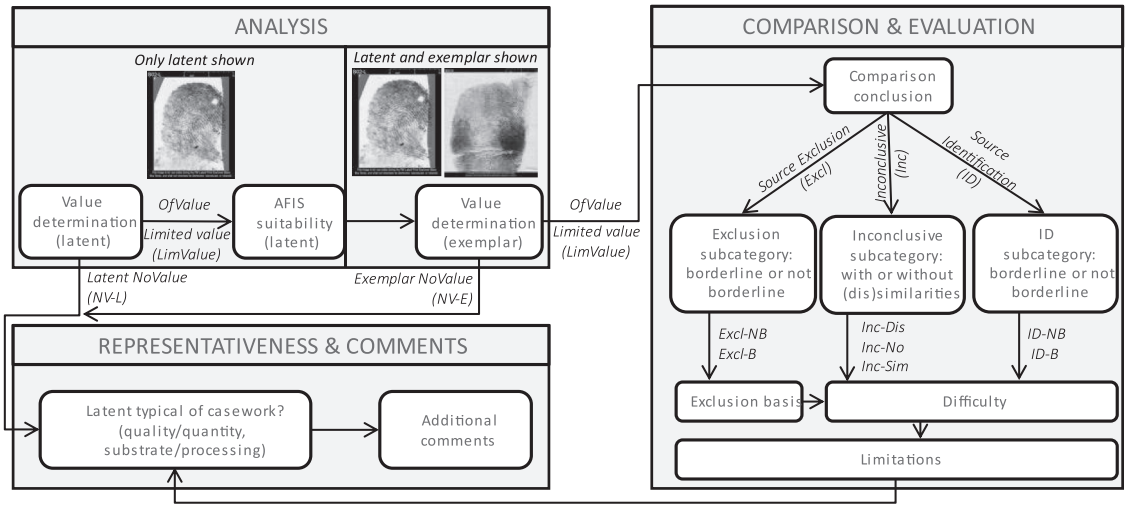


Fig. 1. Study workflow. This process was followed by the software for each image pair assigned to a participant in the study.

2.2. Participation

Because this study was focused specifically on examiner performance on comparisons resulting from NGI searches, participation was limited to LPEs with access to NGI. Participation in this study was open to LPEs who had conducted casework (latent examination, technical review, or verification) as an employee or contractor for a U.S. federal, state, local, tribal, and territorial government agency (i.e., an agency that has access to NGI) within the 2 years prior to study registration.

Participation was solicited at relevant conferences and via professional organization announcements, email communications, and direct requests to laboratory management. A total of 156 LPEs participated, with 126 completing all assigned comparisons (one of whom had two omitted trials (resulting in 98 responses), and one of whom had a few responses submitted under a duplicate registration (resulting in 106 distinct responses)). We do not include in the count of participants the additional 28 examiners who did not complete at least 25 IPs: potential participants were informed they needed to complete at least 25 IPs to be included in analyses. We have learned from previous studies that it is unrealistic to expect all (volunteer) participants to complete all assignments: the study was designed to accommodate by controlling assignments so that each participant who received at least 25 IPs had samples comparable to the overall pool of samples (detailed in Appendix B3.5).

Participants were required to complete an online background questionnaire prior to beginning the study and a brief post-test questionnaire about similarity to casework after completing all assigned comparisons. Of the 156 participants, 79 % work for accredited agencies; 26 % work for U.S. Federal government agencies, 33 % for U.S. State government agencies, and 41 % for U.S. City/County government agencies; 68 % had a formal program of latent print instruction for at least 1 year; 51 % were certified as International Association for Identification (IAI) Certified Latent Print Examiners (CLPEs), 8 % had never been certified or qualified as a LPE, and the remainder had other certifications/qualifications, generally by employers; 89 % had taken a proficiency test in the previous 12 months; 6 % had never taken a proficiency test. A more detailed description of participants can be found in Appendix B2; the complete responses to the background questionnaires (including registrants who did not complete the minimum 25 IPs) can be found in Appendix M.

2.3. Fingerprint data

The fingerprint data included 150 latents (from 150 distinct fingers of 23 different subjects). In general, each of the latents was combined with one mated and one nonmated exemplar to form one mated latent-exemplar image pair (IP) and one nonmated IP for each latent:

Nonmated exemplars were the highest-ranking[2] nonmated exemplar candidate fingerprint returned for the latent searched against NGI.

Mated exemplars were the highest-ranking mated exemplar candidate fingerprint returned for the latent, unless the mate was not included in the top 20 candidates returned by NGI, in which case a mated exemplar on file was selected.

There was one exception to this: due to a file naming error, one latent was inadvertently included in two nonmated IPs (instead of one mated and one nonmated). The study therefore included 151 nonmated IPs and 149 mated IPs. Note that mating is based on ground-truth source attribution: a mated IP refers to a latent and exemplar known *a priori* (through non-fingerprint means) to derive from the same subject and finger; a nonmated IP refers to a latent and exemplar known *a priori* to derive from different subjects. Mating is not derived from examiner (or study team) assessments of IPs: see *Section 3.2* for discussion and results related to LPE consensus.

Latents were selected to be as broadly representative of operational casework as feasible, including a range of processing techniques, substrates, and qualities. A preliminary pool of 206 latents were selected from over 2000 available latents, and searched against NGI; seven of those latents were rejected by NGI due to poor quality or low minutia count. One measure of AFIS performance is the "hit rate," the proportion of searches that return a mated exemplar in the candidate list (which in this case contains 20 candidates). The hit rate of the 199 latents that were searched against NGI corresponded well with the LQMetric [25] latent print quality metric, which returns an estimate of the probability that a search of NGI would hit at rank one if the subject's exemplars are enrolled in the NGI database [25]. For the 199 latents, 106 (53 %) hit at any rank, of which 92 (46 %) hit at rank 1, and 14 (7 %) hit at ranks 2–20. Overall, the average LQMetric value was 50, which can be seen as a reasonable prediction of the 46 % that hit at rank 1. (See Appendix B3.3 for details regarding NGI hit rate with respect to LQMetric.) The primary criterion in down-selecting from the 199 remaining latents to the final 150 latents was making the quality distribution of the associated mated and nonmated exemplars as similar as possible. See Appendix B3 for more details about image selection, processing, and assignments.

All images were cropped close to the fingerprints to remove irrelevant background. Latents that were not obviously upright were presented in an arbitrary orientation. Because images differed in size, a scale was included for each image to avoid ambiguity regarding relative

[2] Note that in this context "highest ranking" refers to a lower number on the candidate list: the rank-1 candidate is higher ranking than the rank-2 candidate.

size. In addition to the unmodified Baseline image, four predefined image transformations were provided for each image. See Appendix B3.4 for additional formatting details.

3. Results

Table 1 defines the datasets used in analyses. Note that the *Baseline Dataset* is used in most analyses: the *Examiner Comparison Dataset* is used for measuring rates for individual participants (*Sections 3.4–3.5*); the *Reproducibility Dataset* is used for measuring the reproducibility of decisions (*Section 3.3*).

For each participant, 20 % of assigned IPs were mated and 80 % were nonmated; these proportions were used in the study design for better measurement of erroneous ID rates. Each mated IP in the *Baseline Dataset* received responses from 15 to 23 participants (mean 19.0, median 19); each nonmated IP received responses from 16 to 82 participants (mean 75.5, median 75). Each participant was assigned 19–20 of the 149 mated IPs (~13 %) and 80–81 of the 151 nonmated IPs (~53 %). The 156 participants each submitted 25–106 responses (mean 91.2, median 100): one participant submitted a few responses under a second registration, resulting in 106 distinct responses; 125 participants completed 100 + IPs (124 participants completed 100 IPs, one completed 106 IPs); 16 participants submitted 50–98 IPs; 15 participants submitted 25–46 IPs.

See Appendix D for further information regarding test yield, omissions, modifications, and anomalies.

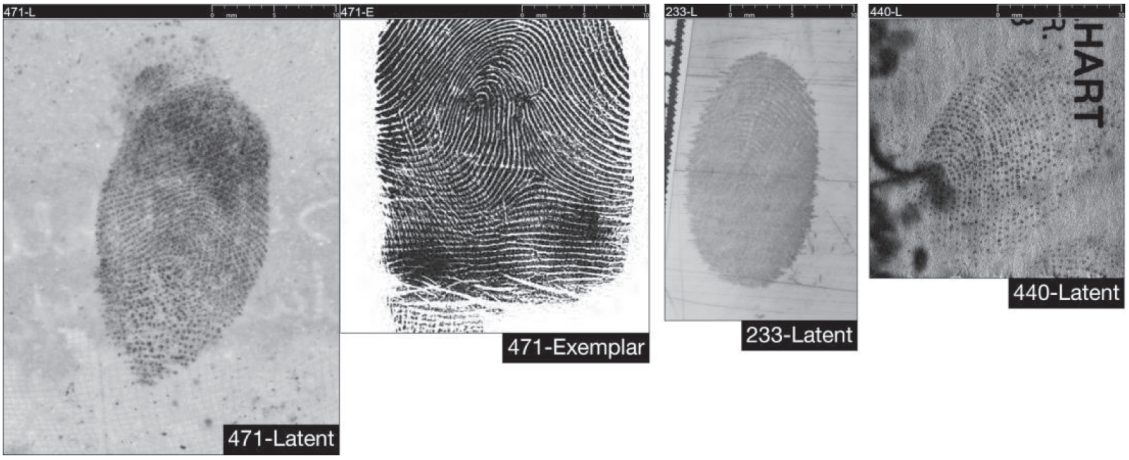
3.1. Decision rates, accuracy, and errors

Fig. 3 shows the distribution of decisions, showing three methods for calculating decision rates when using multi-level conclusion scales. As presented in [7], recommended by the OSAC [26], and used in other studies [14,15], decision rates on multi-level conclusion scales can be reported in several ways, including "PRES," which includes all presentations; "COMP," which includes all comparisons, excluding *NoValue* trials, and "CALLS," which includes only comparisons with conclusive calls (*ID* and *Excl*), excluding trials with *NoValue* or *Inc* responses; see Appendix E for details.

The performance of forensic examiners can be assessed in terms of accuracy (making decisions consistent with ground truth), error (making decisions contrary to ground truth), and effectiveness (making conclusive rather than indeterminate decisions). In terms of accuracy (decisions consistent with ground truth), Fig. 3 shows that the method used can have a notable effect on the rates: the proportion of mated IPs resulting in *IDs* we describe as "true positive rate" (TPR), which varies significantly depending on the method used (TPR_{PRES}=62.6 %, TPR_{COMP}=74.3 %, TPR_{CALLS}=93.8 %); similarly, the proportion of nonmated IPs resulting in *Excls* we describe as "true negative rate" (TNR_{PRES}=69.8 %, TNR_{COMP}=84.2 %, TNR_{CALLS}=99.7 %). In terms of errors (decisions contrary to ground truth), the proportion of nonmated IPs resulting in erroneous *IDs* we describe as "false positive rate" (FPR_{PRES}=0.20 %, FPR_{COMP}=0.24 %, FPR_{CALLS}=0.29 %); the proportion of mated IPs resulting in erroneous *Excls* we describe as "false negative rate" (FNR_{PRES}=4.2 %, FNR_{COMP}=5.0 %, FNR_{CALLS}=6.2 %). The concept of conclusions contrary to ground truth should not be conflated with non-consensus decisions, which we discuss in *Section 3.2*. In terms of effectiveness (making conclusive rather than indeterminate decisions), the rates of indeterminate decisions are of notable importance in understanding the examination process: laboratories need to understand the proportion of latents that do not result in comparison (mated and nonmated *NoValue* rate, $MNVR_{PRES}$=15.8 %, $NNVR_{PRES}$=17.2 %), and the proportion of comparisons that do not result in conclusive decisions (mated and nonmated *Inc* rate, $MIncR_{COMP}$=20.7 %, $NIncR_{COMP}$=15.6 %).

Distinguishing among the PRES, COMP, and CALLS approaches is most intuitive if these are put into words. Using erroneous *IDs* as an example, FPR_{PRES} could be expressed as "If an LPE was presented with a latent and an exemplar from a different person, they made erroneous IDs 0.20 % of the time," FPR_{COMP} as "If an LPE compared a latent that they had assessed to be *VCMP* to an exemplar from a different person, they made erroneous *IDs* 0.24 % of the time," and FPR_{CALLS} as "If an LPE made a conclusive decision when comparing a latent with an exemplar from a different person, they made erroneous *IDs* 0.29 % of the time." Please note that none of these approaches refer to the proportion of IDs that are incorrect: FPR_{CALLS} is easy to misinterpret as "If an LPE made an *ID* decision, 0.3 % of their *IDs* were erroneous," which is a conditional value known as "false discovery rate," discussed below.

As mentioned in Section 2.1, a draft standard latent conclusion scale with a greater number of levels is currently under consideration [24]. The Texas Forensic Science Commission recently made the



IP	Mating	Total	Excl-NB	Excl-B	Inc-Dis	Inc-No	Inc-Sim	ID-B	ID-NB	NV-L
471	Mated	17	7	2	1	1	3	2	1	
233	Nonmated	80	10	9	3	4	10		1	43
440	Nonmated	80	46	20	1	3	4		1	5

Fig. 2. Examples of image pairs (IPs). IP 471 had the highest rates of erroneous exclusions in the study. None of the exemplars from nonmated IPs can be shared publicly (see Appendix B3 for discussion). IPs 233 and 440 each had one erroneous ID and at least three Inc-Sims (five other IPs in the study also had one erroneous ID and three Inc-Sims). All images are unmodified Baseline images, shown at the same resolution.

Table 1
Datasets used in analyses.

Dataset	*N*				*Distinct IPs*			*Participants*	*Description*
	Type	*Total*	*Mated*	*Nonmated*	*Total*	*Mated*	*Nonmated*		
Baseline Dataset	(Responses)	14,224	2830	11,394	300	149	151	156	Default dataset for analyses
Examiner Comparison Dataset	(Responses)	13,695	2729	10,966	300	149	151	141	Subset of *Baseline Dataset* for measuring individual rates and comparing examiners. Omits 15 participants who did not complete at least 50 IPs each.
Reproducibility Dataset	(Decision pairs)	904,692	51,588	853,104	300	149	151	156	904,692 interexaminer decision pairs derived from a self-join of the 14,224 responses in the *Baseline Dataset* (each individual response is paired with every other response on the same IPs)

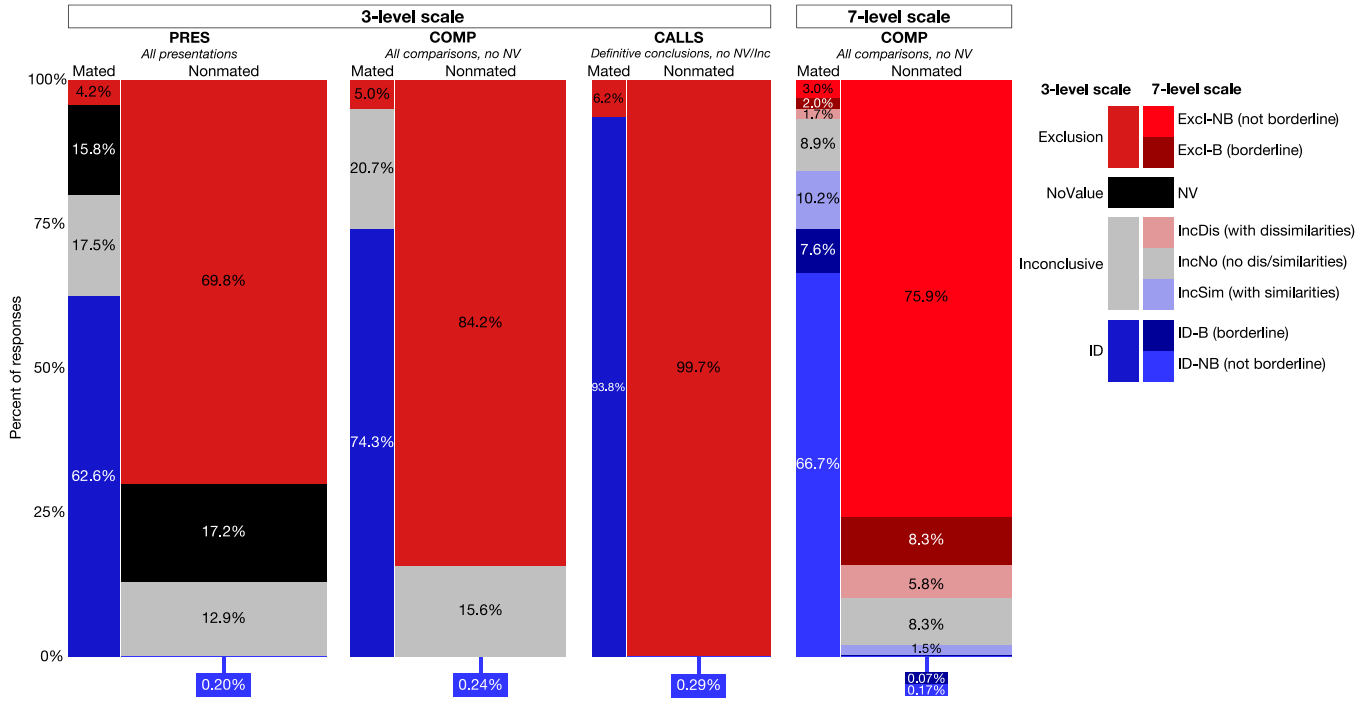


Fig. 3. Mosaic plots of the distribution of decisions on mated and nonmated IPs. The 3-level traditional scale is shown using the PRES, COMP, and CALLS methods for calculating decision rates, compared against the 7-level scale using COMP rates. The 7-level scale uses the COMP method for calculating decision rates, since NoValue responses do not differ between the 3- and 7-level scales. Column widths are proportional to counts. (PRES: 2830 mated and 11,394 nonmated responses. COMP: 2383 mated and 9439 nonmated responses. CALLS: 1889 mated and 7971 nonmated responses). See Appendix E for counts, confidence intervals, and further discussion. For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.

recommendation "FSSPs [forensic science service providers] should consider incorporating examination conclusions that expand on the traditional three-tiered decision categories (source identification, inconclusive, source exclusion) to a more nuanced decision expression." [27] Fig. 3 compares the three-level traditional scale against the seven-level scale resulting from subcategories of *ID*, *Inc*, and *Excl* responses. This information may be of value to agencies considering changing conclusion scales, indicating how different scales correspond. Note that the borderline and inconclusive/leaning categories are relatively symmetric: nonmated *Excl-B* and *Inc-Dis* are both ~6–8 %, mated *Excl-B* and *Inc-Dis* are both ~2 %, and mated *ID-B* and *Inc-Sim* are both ~8–10 %; this does not hold true for nonmated *ID-B* and *Inc-Sim*, presumably because nonmated *IDs* are such rare occurrences.

Overall decision rates provide an overall assessment of the participants' performance. However, such rates are an average over all participants and can be disproportionately affected if any participants have individual rates that are strikingly different from most participants. This study had one such outlier participant: of the 23 erroneous *IDs* (FPs), 13 were made by a single participant (discussed further in *Section 3.4*). The notable variability in examiner performance underscores the extent to which rates of rare events (such as erroneous *IDs*) are sensitive to individual participants: the FPR for this study would be nearly identical to LatentBB09 if the outlier participant were not included. Bootstrap sampling allows us to see those effects, as a distribution of estimates of rates: each chart in Fig. 4 is calculated by selecting 156 subjects at random from the study participants, with replacement (i.e., a study participant may be included multiple times or not at all), calculating the resulting rates, and repeating the process for 5000 iterations. Fig. 4 shows that for TPR_{PRES}, TNR_{PRES}, and FNR_{PRES}, the bootstrapped rates are not very sensitive to the individual participants. However, the bootstrapped FPR_{PRES} rates show us that FPR_{PRES} is very dependent on the individual participants: the peaks show the effect of having an examiner like the outlier examiner included in the 156 participants [0,1,2,3,4] times; the differing heights of the troughs between the peaks show the effects of including varied numbers of the other participants with FPs. These results show the robustness of the measured rates with respect to individual participants: the rates other than FPR are relatively stable with respect to the individual examiners (limited to the pool of participants), but FPR is highly sensitive to individual variation. (Note that out of the 23 FPs, the outlier participant made 57 %; of the 118 FNs,

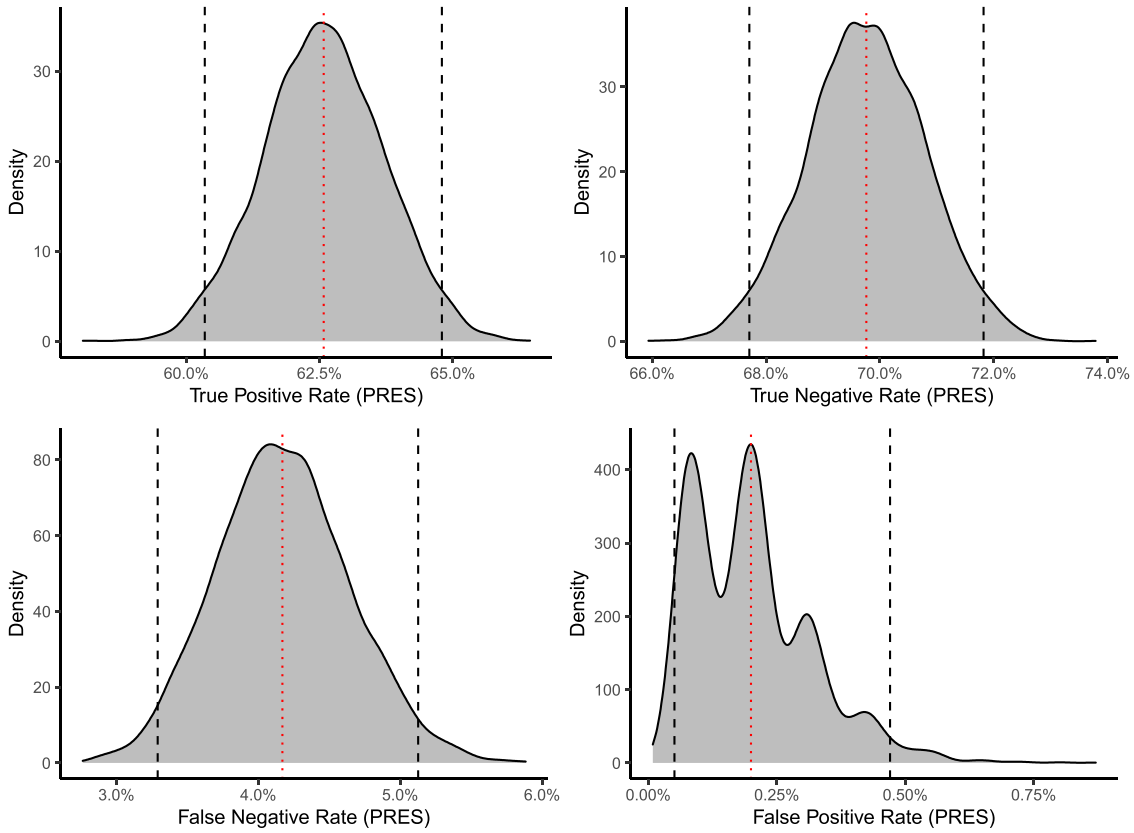


Fig. 4. Probability density function distributions of PRES decision rates using bootstrap sampling, conducted using 5000 iterations, sampling 156 participants with replacement. Dotted red line: rate as measured in LatentBB22. Dashed black lines: 95 % confidence bounds on distribution of rate (at 2.5 % and 97.5 % of the 5000 iterations).

no participant made more than 5 %; of the thousands of TNs and TPs, no participant made more than 1 %.) See Appendix E1 for confidence intervals, calculated using both bootstrapping and Clopper-Pearson approaches.

The proportions of each decision (PRES, COMP, and CALLS rates for TPR, FPR, etc.) are important accuracy measures, but assume *a priori* knowledge of true mating relationships, which of course are not known in forensic casework. In practice, it is far more important to know the likelihood that a given decision is correct: the probability that an *ID* is on a mated image pair (also known as positive predictive value, PPV), or

Table 2
Decision counts and conditional values. Rescaled proportions convert the PRES proportions (19.9 % mated, 80.1 % nonmated) to (50 % mated, 50 % nonmated). Yellow highlights indicate errors; blue highlights indicate Inc-Dis and Inc-Sim responses that are contrary to ground truth. Bold outlines indicate the rescaled false omission rate (FOR), negative predictive value (NPV), positive predictive value (PPV), and false discovery rate (FDR); see text for discussion.

			Total	*Mated*	*Nonmated*	*% Mated (As measured, 19.9% mated)*	*% Nonmated (As measured, 19.9% mated)*	*% Mated (Rescaled to 50% mated)*	*% Nonmated (Rescaled to 50% mated)*
No Value	NV (Exemplar)		15	5	10	33.3%	66.7%	66.8%	33.2%
	NV (Latent)		2,387	442	1,945	18.5%	81.5%	47.8%	52.2%
	NV (total)		2,402	447	1,955	18.6%	81.4%	47.9%	52.1%
3-level	Excl		8,066	118	7,948	1.5%	98.5%	5.6%	94.4%
	Inc		1,962	494	1,468	25.2%	74.8%	57.5%	42.5%
	ID		1,794	1,771	23	98.7%	1.3%	99.7%	0.3%
7-level	Excl	Excl-NB	7,232	71	7,161	1.0%	99.0%	3.8%	96.2%
		Excl-B	834	47	787	5.6%	94.4%	19.4%	80.6%
	Inc	IncDis	583	40	543	6.9%	93.1%	22.9%	77.1%
		IncNo	992	212	780	21.4%	78.6%	52.2%	47.8%
		IncSim	387	242	145	62.5%	37.5%	87.0%	13.0%
	ID	ID-B	188	181	7	96.3%	3.7%	99.0%	1.0%
		ID-NB	1,606	1,590	16	99.0%	1.0%	99.8%	0.2%
Total Presentations (PRES)			14,224	2,830	11,394	19.9%	80.1%	50.0%	50.0%
Total Comparisons (COMP)			11,822	2,383	9,439	20.2%	79.8%	50.4%	49.6%
Total Calls (CALLS)			9,860	1,889	7,971	19.2%	80.8%	48.8%	51.2%

the probability that an *Excl* is on a nonmated image pair (also known as negative predictive value, NPV); these can also be expressed as the likelihood that a given decision is erroneous (1-PPV is the false discovery rate (FDR), and 1-NPV is the false omission rate (FOR)). These conditional values are posterior probabilities, which are functions of the proportions of mates vs. nonmates: in Table 2 we rescale the measured proportions (based on 19.9 % mates vs. 80.1 % nonmates) to assume equal proportions of mates and nonmates (i.e., 50 % mates and 50 % nonmates) to serve as a neutral baseline. In casework, the prior prevalence of mates vs. nonmates would vary substantially among organizations, by case type, and by how candidates are selected. Table 2 shows that the less conclusive decisions have lower conditional values: the rightmost column shows us that for the continuum from *Excl* to *Inc*, *Excl-NB*= 96.2 % nonmated, *Excl-B*= 80.6 %, and *Inc-Dis*= 77.1 %. For the continuum from *ID* to *Inc*, *ID-NB*= 99.8 % mated, *ID-B*= 99.0 %, and *Inc-Sim*= 87.0 %.

Note that an ~20 % mate proportion is a reasonable estimate that a latent from an arbitrary person in the U.S. would have the mate returned as the rank-1 candidate as the result of an NGI search. When the LatentBB22 image pairs were selected in 2021, NGI contained approximately 128 million subjects, and the U.S. population was approximately 332 million (38.6 % of the population was in NGI). Over a dataset of latents of varied quality such as those in this study, a latent could be expected to average about a 50 % probability of resulting in a rank-1 hit if the subject is in NGI (see "Results with respect to latent quality" in Section 3.7). Therefore, a latent from an arbitrary person in the U.S. is about 19 % likely to have the mate returned as the rank-1 candidate. Note however that the 19 % likelihood is for an *arbitrary* person, and the probability that a rank-1 candidate would be a mate would increase for subjects who are more likely to be in NGI (e.g., if they are more likely to have a criminal or civil record).

3.2. Consensus

Each mated IP was examined by an average of 19.0 participants, and each nonmated IP was examined by an average of 75.5 participants. Their collective decisions can be considered votes in a decision space, as shown in Fig. 5: for each IP, this shows the percent of examiners who determined there was sufficient information for an *ID* or *Excl* decision. No nonmated IP received more than one FP, which can be seen in Fig. 5 as the (red) nonmated pairs that are just to the right of the vertical axis; since each of these received one erroneous ID out of 72–80 total responses, there is very little horizontal spread. Conversely, the prevalence of FNs can be seen in the vertical spread of *Excls* on the (blue) mated IPs, many of which received more than one *Excl*, and two of which received a majority of *Excls* (FNs). With respect to decisions consistent with ground truth, the proportion of *IDs* on each mated IP ranged from 0 % to 100 % (TPR_{PRES}): 27 mated IPs (18 %) yielded no *IDs*, and on 95 (64 %) the majority of responses were *IDs*. For the nonmated IPs, the proportion of *Excls* per IP also ranged from 0 % to 100 % (TNR_{PRES}); only 3 nonmated

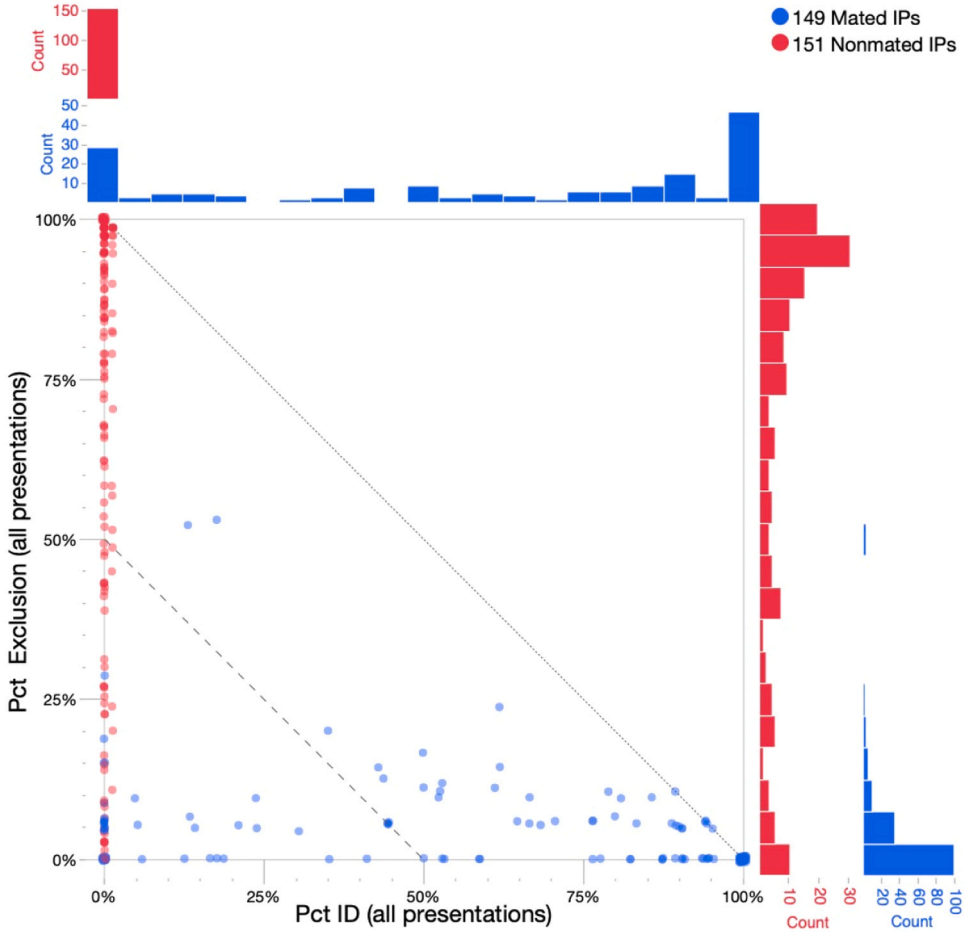


Fig. 5. Decision rates on each image pair (IP), showing the percentage of participants making an ID decision (x axis) vs. Excl decision (y axis) on each IP; mean 19.0 presentations per mated IP, 75.5 presentations per nonmated IP. NoValue decisions are treated as inconclusive. Marginal distributions are shown as histograms. Points are jittered to minimize superimposition. Superimposed points in the corners: of mated IPs, 46 (31 %) were unanimous IDs (true positives), 13 (9 %) unanimous IncNV; of nonmated IPs, 19 (13 %) were unanimous Excls (true negatives), 3 (9 %) were unanimous IncNV. Points along the main diagonal represent IPs on which all examiners reached Excl or ID conclusions. Points below the dashed diagonal represent IPs on which the majority of participants reached IncNV decisions. For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.

IPs (2 %) yielded no *Excls*, and on 109 (72 %) the majority of responses were *Excls*.

Fig. 5 can be used to understand the extent of consensus for each IP. For the IPs between the two diagonals, the majority of participants made a conclusive response (*ID* or *Excl*), as opposed to an indeterminate response (*Inc* or *NoValue*); between the origin and the dashed diagonal, the majority of participants made *Inc* or *NoValue* decisions. Assessing consensus (sometimes described as the "forensically appropriate" decision) is not necessarily as simple as determining the majority response: consensus could also be assessed using median, average, or plurality responses, or the interquartile range (see Appendix F for discussion and additional details) — and these methods often differ in what would be assessed as the consensus decision. In making comparison determinations, an expert assesses whether there is sufficient data to support and justify a given decision, as we have explored in detail previously [10,28,29]. This assessment does not yield a single, fixed threshold, but rather a continuum between indeterminate and conclusive responses; Fig. 5 shows this continuum as an aggregate of responses among examiners. Other than for contradictory decisions (*Excl* vs. *ID*), differences in decisions are disagreements regarding whether there is sufficient information to make a given decision. For example, an ID decision is not just an assessment that a latent was from the same source as a given exemplar, but an assessment that there is sufficient quality and quantity of information in a comparison of the latent and exemplar to conclude that they came from the same source, and therefore an *ID* vs. *Inc* disagreement relates to different assessments of sufficiency. The consensus-based view of the results shown in Fig. 5 provides an illustration of collective votes on sufficiency. The 23 nonmated IPs (red points) just to the right of the vertical axis each received one erroneous *ID* out of 72–80 responses: the chart provides a visual depiction that whatever information those participants used in making their decisions, none of the other 71–79 participants who received those IPs considered that there was sufficient information to make an *ID*. Contrast this with the vertical spread of mated IPs (blue points), 25 of which had at least two participants making erroneous *Excls*, incorrectly corroborating their assessments of sufficiency to exclude.

When experts disagree, it is an oversimplification to say that they can or should be judged as correct or incorrect: decisions that contradict known ground truth are demonstrably wrong and should be considered errors (such as an *ID* on a nonmated IP), but it is problematic for decisions that disagree with a majority of other examiners with respect to sufficiency to be judged as strictly right or wrong (as suggested by [30]; see [31] for discussion). Such non-consensus decisions [32] may reflect differences in risk tolerance, confidence, or skill level. For example, given an IP on which 75 % of examiners make a conclusive response, it would be tempting to judge the 25 % who made indeterminate responses as incorrect, but these different assessments of sufficiency could instead be the effects of different levels of caution or risk tolerance. Note that conclusive decisions may also be non-consensus decisions: if one examiner out of 50 makes an *ID* decision on a mated IP (and the other 49 make indeterminate decisions), it may be evidence of an unusually high level of skill or an unusually high risk tolerance, and only further testing would be able to differentiate. With those caveats, we note that 4.5 % of *IDs*, 9.0 % of *Excls*, and 25.9 % of *IncNVs* differed from the majority response for each IP (see Appendix F2 for discussion and additional details).

3.3. Reproducibility

Interexaminer variability can be described as "system noise," which Kahneman, Sibony, and Sunstein define as "unwanted variability in judgements that should ideally be identical" [33]. Consensus and reproducibility are two complementary methods of evaluating interexaminer variability, both of which are assessments of the extent of agreement among examiners on the same IPs. In our usage, "consensus" refers to the extent to which examiners concur on their decisions for each IP (as shown in the previous section), whereas the similar concept "reproducibility" refers to how frequently different participants reach the same decisions when each response for a given IP is compared to all other responses for that IP. We use both approaches as they address different purposes: consensus provides an IP-centric means of assessing variability, whereas reproducibility assesses variability across all IPs [33].

To assess reproducibility, we use the *Reproducibility Dataset*, which is a self-join of the *Baseline Dataset*: each individual response is paired with every other response on the same IPs, resulting in 904,692 interexaminer decision pairs derived from the 14,224 responses in the *Baseline Dataset*. In terms of reproducibility of latent value assessments when using the [*NoValue, LimValue, OfValue*] response categories, when two participants received the same latent, 74.3 % of suitability determinations agreed exactly, and 7.0 % were contradictory (*NoValue* vs. *OfValue*); 18.7 % shifted one category (*NoValue* vs. *LimValue*, or *OfValue* vs. *LimValue*). If we combine *LimValue* and *OfValue* (reducing the value assessment to be Boolean), participants agreed 85.9 % and disagreed 14.1 % of the time. (See Appendix G1 for details.)

In terms of reproducibility of comparison decisions when using the [*ID, IncNV, Excl*] response categories (not differentiating between *Inc* and *NoValue*), when two participants received the same IP, 79.1 % of decisions agreed exactly, 20.4 % disagreed regarding sufficiency (*ID* vs. *IncNV*, or *Excl* vs. *IncNV*), and 0.5 % were contradictory (*Excl* vs. *ID*). The reproducibility of comparison decisions is strongly associated with the participants' assessments of difficulty. For example, on trials a participant assessed as *VEasy*, 97.4 % of other participants made the same comparison decision, but on trials a participant assessed as *VDiff*, only 47.9 % of other participants made the same comparison decision. Overall, almost all of the non-reproduced decisions were disagreements regarding sufficiency, in which one participant made a conclusive decision (*ID* or *Excl*), and another made an indeterminate decision (*Inc* or *NoValue*) — and disagreements regarding sufficiency increased notably with increased difficulty. Conclusive decisions consistent with ground truth have comparable rates of agreement (85 % of TNs and 87 % of TPs were reproduced). Most errors were not reproduced: FP errors were never reproduced, but 15 % of FN errors were reproduced. (See Appendix G2 for details)

The psychology literature has shown that the number of categories in conclusion scales has notable effects on the measurement of reproducibility (see e.g. [34,35]). The results of this study corroborate our findings in [28], in which we stated "When two examiners reached *different* conclusions on the three-level scale, the extent of disagreement was often *over*stated, because either or both of the examiners' decisions were borderline on the seven-level scale. Conversely, when two examiners reached the *same* conclusion on the three-level scale, the extent of disagreement was often *under*stated, because either or both of the examiners' decisions were borderline on the seven-level scale." (See Appendix G3 for details.)

3.4. Comparing participants' performance

Examiner performance for latent comparison decisions is multifaceted. When using multi-level conclusion scales, performance evaluations must consider both accuracy (i.e., correctness) and indeterminacy (i.e., lack of definitiveness), which together are an indicator of an individual LPE's effectiveness. A highly effective examiner would have both high rates of accuracy and high definitiveness—they would often report *ID* or *Excl,* and those decisions would be correct. Although sample characteristics (e.g., latent quality, comparison difficulty) can have an impact on effectiveness in individual comparisons, overall effectiveness (across a number of comparisons with a range of attributes) is a function of examiner tendencies such as skill and risk aversion.

Fig. 6 compares the participants in terms of accuracy, indeterminacy, and effectiveness for each participant in the *Examiner Comparison Dataset*. Note that when measuring individual participant rates and

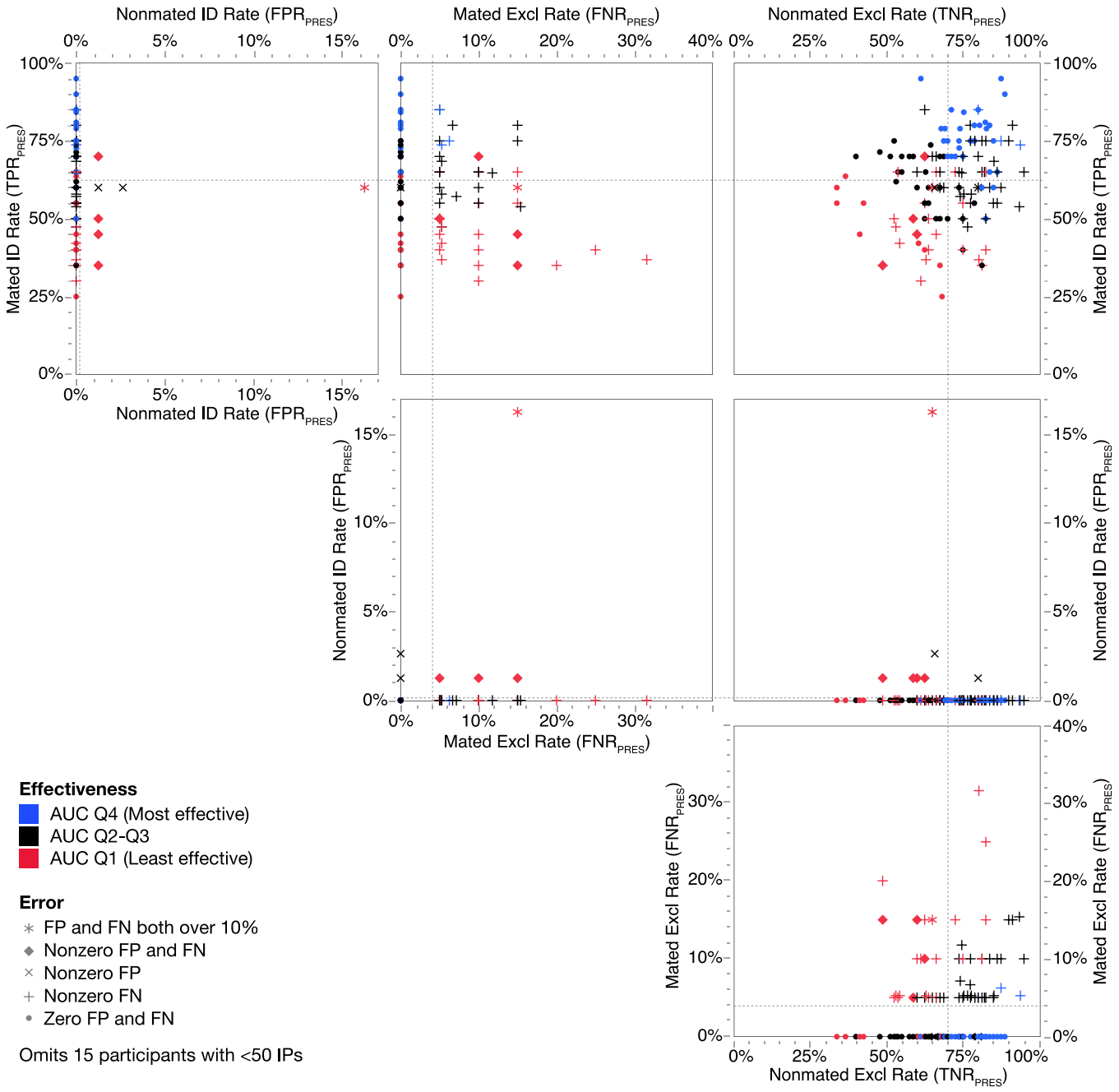


Fig. 6. Scatterplot matrix comparing participants in terms of performance. Color-coding depicts effectiveness as measured by AUC quartiles; symbols indicate extent of errors. Means for each rate are shown as dotted lines. Markers are jittered to minimize superimposition. For these results with the addition of IncNV rates, see Appendix H1. (Examiner Comparison Dataset. n = 141 participants who completed at least 50 assigned IP examinations; rates calculated from 13,695 responses.). For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.

comparing examiners, we limit our analyses to the 141 participants who completed at least 50 % of their assigned comparisons (*Examiner Comparison Dataset*). Each panel in Fig. 6 contrasts two different decision rates, wherein each point represents an individual participant, each of whom is represented in all panels; the colors of each point denote the participant's overall effectiveness. We assess participants' effectiveness in terms of the area under the curve (AUC) of a receiver operating characteristic (ROC) curve (similar to the manner in which it was done in [16]). This approach uses the participants' responses as predictors of ground truth: we model two decision thresholds by grouping indeterminate responses (*IncNV*) first with *IDs*, then with *Excls* (see Appendix H2 for additional details). The most effective participants (those in the upper AUC quartile) in this study are shown in blue in Fig. 6, whereas the least effective participants (those in the lower AUC quartile) are shown in red, and the intermediate participants are shown in black. For example, in the left-most panel of the second row in Fig. 6, the red asterisk represents the outlier participant who had the highest FPR (16 %), and this participant also had an above-average FNR (15 %); following that point to the other panels, we see that participant also had a slightly below average TPR (60 %) and TNR (65 %), likely due to the number of erroneous decisions reported. However, this participant was more definitive than average (20 % *IncNV* Rate); in other words, this participant reported fewer determinations of *IncNV* than average, but this came at the expense of higher error rates. All together, these reporting tendencies result in this participant's effectiveness falling in the lowest quartile.

In general, Fig. 6 shows that high AUCs (i.e., high efficacy) result from not only high rates of correct decisions, but also low rates of both erroneous decisions and indeterminate responses. The most effective participants (blue circles) are generally above average for both TPR and TNR and have no erroneous decisions, and below average *IncNV* rates. For additional details regarding accuracy and effectiveness, as well as AUC results on an expanded 7-level conclusion scale, see Appendix H.

Accuracy and error rates were not equally distributed among participants. Erroneous *Excl* decisions were not limited to a few individual participants: 72 of the 156 participants reported at least one FN, and three participants made 4–6 FNs each. However, only ten (out of 156) of the participants in this study made any erroneous *IDs*: eight participants made one, one participant made two, and, as previously discussed, one outlier participant made 13 such errors.

The outlier participant (shown as the red asterisk in Fig. 6) is currently a LPE at a local laboratory, their laboratory was either unaccredited or they were unsure whether their laboratory was accredited, has 1–4 years of experience, has never testified in court as an LPE, had limited formal training (restricted to courses and workshops only), is not IAI certified, has never been notified that they have made a FP error after training, and uses the same decision criteria when making *IDs* for AFIS and non-AFIS casework (discussed in *Section 3.5*)— including the outlier participant, a total of five participants had these same attributes. Of the ten participants who made erroneous IDs in the study, only half (five) had ever taken a proficiency test; of the 146 other participants, 97 % (142) had taken proficiency tests.

3.5. Associations between examiner background and performance

To further characterize the performance of LPEs in this study, we utilized a multi-phasic approach for detecting and reporting any associations between various attributes collected in the background questionnaire and performance. In particular, we evaluated the performance of 140 participants (from the *Examiner Comparison Dataset*, omitting the outlier participant because their anomalous performance would inappropriately skew the results) with respect to 16 attributes (regarding their latent training, experience, and casework/laboratory policies), considering six metrics that capture accuracy, definitiveness, and effectiveness (TPR_{PRES}, TNR_{PRES}, FPR_{PRES}, FNR_{PRES}, IncNV Rate, and AUC). To do so, we used two complementary approaches for detecting and reporting any such associations: variable importance analysis (VIA) and attribute-specific significance testing, as developed and described in [16] and used in [14,15,36]. See Appendix I1 for additional details.

For the majority of attributes that we evaluated—including years of latent experience, comparison frequency, knowledge of previously reported errors, and frequency of AFIS/NGI usage—we did not detect support for an association with any of the six performance measures. We did, however, detect *notable support for associations* between performance and four attributes:

- ***Accreditation*** — participants who worked at an accredited agency had significantly higher TPR_{PRES}, lower FNR_{PRES}, and accordingly higher effectiveness (AUCs) overall than those who did not (or those who were unsure whether their laboratory was accredited).
- ***Use of a minimum # of minutiae threshold for AFIS searches*** — participants who did not use such a threshold (either formally or informally) had significantly lower TNR_{PRES} and higher IncNV Rates than those who used a formal threshold, either as specified by their SOPs or as required by their AFIS.
- ***Training program*** — participants who completed a formal training program lasting 1 year or longer had significantly lower FNR_{PRES} than participants who completed informal/limited formal training (e.g., just courses and workshops).
- ***Proficiency testing*** — participants who had never completed a proficiency test were more likely to exhibit higher rates of erroneous *IDs* than those who had completed a proficiency test, irrespective of whether they had taken that test in the last year or not. Because FPs were extremely sparse in this study (only 7 out of 13,595 decisions reported by the 140 participants included in this analysis), we caution the interpretation of this result; this effect may or may not persist with additional data collection.

Note that in practice each of these four attributes are associated with agencies (or their policies), rather than individual examiners. With respect to the participants in this study, we found that all four attributes exhibited moderate to strong correlations with employer based upon a bias-corrected Cramer's V analysis [37,38]. More specifically, participants from City/County agencies were much less likely than those from Federal and State agencies to work for an accredited agency, have completed a formal training program lasting a year or longer, or completed a proficiency test within the last year; they were also much less likely than participants from Federal agencies to use a formal minutiae threshold for AFIS searches. Participants from City/County agencies made a disproportionate number of the errors reported in this study, which may be explained (at least in part) by the fact that these participants were much more likely to have never taken a proficiency test or work for an unaccredited agency. Omitting the outlier participant, 39 % of responses were reported by participants employed by City/County agencies, but they reported 90 % (9/10) of the FPs and 56 % (64/115) of the FNs. City/County examiners were also less likely than State or Federal examiners to be in the "most effective" quartile (blue in Fig. 6).

For additional discussion of these results, details about the three attributes that exhibited limited support for associations with performance, and insights into the correlations between each of the background attributes, see Appendix I.

Of the Background Questionnaire questions, one question that warrants additional discussion is Q#26, in which participants were asked "When you are comparing latent prints to exemplars and considering whether to make an ID conclusion, is your decision process affected based on whether the exemplar was the result of an AFIS search vs. the result of non-AFIS investigation?": 4 % indicated their agency has explicit policies; 19 % indicated their agency recommends greater caution; 35 % indicated their agency policies do not address this, but they personally use greater caution; and 40 % indicated their decision process for making *IDs* is the same regardless of whether the exemplar was AFIS vs. non-AFIS. *(2 % of participants had no AFIS casework in the last two years and were not asked the question.)* All ten participants who made FPs work at agencies that do not provide policies or guidance regarding *IDs* in AFIS vs. non-AFIS casework; six of those (including the two participants who made multiple FPs) indicated their decision process for making *IDs* is the same regardless of whether the exemplar was AFIS vs. non-AFIS. See Appendix I4 for discussion and details.

3.6. Comparison with the original black box study

As discussed above, LatentBB22 was based on LatentBB09, but was designed to measure decision rates on fingerprints representative of NGI and incorporated a variety of lessons learned regarding study design, implementation, and data analyses. Appendix J provides a detailed comparison of LatentBB09 and LatentBB22, summarized here:

Although both studies were open to practicing LPEs, LatentBB22 restricted participation to LPEs who had access to NGI. In LatentBB09, 8 % of participants were non-U.S. or from the private sector.

The latents, exemplars, and IPs in LatentBB22 were selected to be representative of NGI casework. In contrast, LatentBB09 was not focused on representing any specific use case: LatentBB09 selected a broad range of latents, exemplars, and IPs similar to those that could be encountered in casework, but not necessarily representative of any particular type of casework.

In LatentBB22, 80 % of assignments were nonmated (vs. an average of 32 % in LatentBB09), for better measurement of the FPR.

In LatentBB22, participants received equivalent tests in terms of mating, and the quality and attributes of latents. In LatentBB09, assignments were random, and the image pairs received were often notably different among participants.

LatentBB09 included reassignments of samples to test repeatability, which was not included in LatentBB22.

The two studies were generally comparable in test yield in terms of numbers of responses and participants: LatentBB09 had 17,121 responses from 169 participants; LatentBB22 had 14,224 responses from 156 participants. The differences in test yield related to the proportions of mates vs. nonmates, and the number of responses per IP: for LatentBB09, IPs had an average of 22.3 responses for mates and 24.7 for nonmates; for LatentBB09, IPs had an average of 19.0 responses for mates and 75.5 for nonmates.

In LatentBB09, the mated IPs had a much greater proportion of very poor-quality latents than nonmated IPs, in order to provide data to help understand *Value* vs. *NoValue* decisions (which we assessed in detail in [29]). This explains why on mated IPs LatentBB09 had a higher latent *NoValue* rate (29.3 % vs. 15.6 %), and (when combined with the fact that in LatentBB09 mated IPs also had a greater proportion of poor-quality exemplars than nonmated IPs) at least partially explains the higher *Inc* rate (33.5 % vs. 17.5 %). These in turn explain the notable differences in TPR_{PRES} (*IDs* on mated IPs: 32.0 % LatentBB09 vs. 62.6 % LatentBB22).

The higher rate of inconclusives in LatentBB09 is interesting in that inconclusive decisions were often discouraged by agency policies when LatentBB09 was conducted. In LatentBB09, 32 % of participants indicated *Inc* decisions were not permitted when the exemplar and latent were both *OfValue* (compared with 5 % in LatentBB22); only 48 % of LatentBB09 participants said *Incs* were freely accepted in such cases (compared with 79 %). After LatentBB09, some laboratories were concerned that LPEs might feel pressure to make conclusive decisions against their professional judgment and changed policies in response.

FNR was higher in LatentBB09, particularly for FNR_{CALLS}: 14.2 % vs. 6.2 %. We believe that this may have been an effect of the results of LatentBB09: after that study was published, the unexpectedly high FNR was widely discussed in the LPE community, and many labs added more training specific to exclusion decisions and revised their SOPs regarding exclusions. We assume but cannot know that these changes in response to LatentBB09 resulted in the lower FNR measured in LatentBB22.

FPR was lower in LatentBB09 (0.1 % vs. 0.2 % FPR_{PRES}; 0.2 % vs. 0.3 % FPR_{CALLS}). As discussed in *Section 3.1*, those differences can be explained completely by one outlier participant. FPR was very dependent on the individual participants, and bootstrapped rates provide a more complete understanding of the rates and their limitations.

Normalized positive predictive values were the same for both studies; negative predictive values were similar.

Reproducibility rates were very similar, after accounting for the higher FNR in LatentBB09.

3.7. Additional results

In addition to the core analyses, various additional assessments were conducted, the most notable of which are summarized here; see Appendix K for details on all additional results.

Results with respect to latent quality — The distribution of decisions is strongly associated with latent quality, as measured by LQMetric. The proportion of NV-L responses increases as LQMetric decreases: 37 % of responses were NV-L on latents of the lowest quality (LQMetric $\leq$ 19), but $<$ 1 % on latents of the highest quality (LQMetric $\geq$ 80). The proportion of inconclusive responses was strongly associated with latent quality for mated IPs: 30 % of responses were *Inc* on the lowest quality latents, but 2 % on the highest quality latents. Interestingly, inconclusive responses were not strongly associated with latent quality for nonmated IPs: 9–16 % of responses were *Inc* without a clear relationship to latent quality. The result of this is that the proportion of conclusive decisions is strongly associated with latent quality, especially for *IDs* on mated IPs. See Appendix K1 for details.

Decision rates by difficulty — In each COMP trial (i.e. not on trials resulting in NV-L or NV-E), the participants assessed the difficulty of the comparison, using a scale of [Very Easy/Obvious, Easy, Moderate, Difficult, Very Difficult]. Decisions were strongly associated with difficulty: as difficulty increases, the proportion of inconclusive and borderline decisions increases. The relationship of difficulty assessments to latent quality depends on the type of decision: for *IDs*, difficulty is strongly associated with latent quality, but there is only a limited association between difficulty and quality for *Excl* or *Inc*. Participants who made erroneous decisions (FP or FN) often assessed them as more difficult than decisions consistent with ground truth (TP or TN). See Appendix K2 for details.

Decision rates by NGI matcher score — Given that a fingerprint matcher is designed to quantify the similarity of two fingerprints, it is of interest to see the extent to which NGI matcher scores correspond to the proportion of *IDs* a given IP receives, as the latter can be seen as votes among experts assessing the extent to which an *ID* decision is justified. The proportion of participants who made *ID* decisions is strongly associated with NGI scores, and the proportion of indeterminate decisions and FNs is inversely related to NGI scores. See Appendix K3 for details.

Assessments of Limited Value — Note that although LatentBB09 and LatentBB22 both use a 3-level value scale, in this study we refer to the intermediate value decision as "limited value" rather than "value for exclusion only" (VEO), which we used in LatentBB09: LatentBB09 results (and additional analyses in [29]) showed that VEO was not a reproducible category and despite the name was used in some cases for identification; "limited value" better represents how the category is used. In this study, *LimValue* responses were far less likely than *OfValue* responses to result in conclusive non-borderline decisions; on mated IPs, the rate of FNs was greater for *LimValue* than *OfValue* responses. Participants did not generally agree on *LimValue* decisions: only 28 % of *LimValue* decisions were reproduced. See Appendix K4 for details.

Assessments of Value for AFIS — For each trial in which the participant assessed the latent *OfValue*, participants were asked if the latent was of suitable quality to search an AFIS. Most participants use such a designation in casework. Participants' assessments of AFIS suitability generally predicted the NGI hit rate: of the latents that a majority of participants assessed as AFIS suitable, 70 % hit in NGI, as compared to 16 % of the latents that were not majority AFIS suitable. AFIS suitability assessments were usually reproduced: overall, participants agreed on AFIS suitability 78.7 % of the time, and disagreed 21.3 % of the time. See Appendix K5 for details.

Missingness — Overall, we had an 8.9 % item nonresponse rate in this study (14,224 responses collected; 15,606 expected). Of the 156 participants, 31 (20 %) did not complete all 100 assignments; these participants were not significantly different from those who completed all 100 comparisons with respect to any of the attributes for which we found notable associations with performance. If analyses were restricted to participants who completed all assignments, the differences in decision rates would be negligible. To account for missingness, we used simple imputation and found that observed rates and the imputed rates were comparable, and their 95 % Clopper-Pearson CIs overlapped for all decision rates (FPR_{PRES}, FNR_{PRES}, TPR_{PRES}, TNR_{PRES}, IncNV-M_{PRES}, IncNV-NM_{PRES}). See Appendix K6 for details.

Representativeness — Participants were asked to assess the representativeness of the latents for each comparison, as well as an overall assessment of both exemplars and latents in the post-test questionnaire. For almost all of the latents in the study, the majority of participants assessed each latent typical in terms of quality, quantity of information, substrate, and processing. When asked about the overall representativeness of the fingerprints in the study, a supermajority of participants (80 %+) assessed the latents and exemplars similar to casework in terms of quality, and the IPs typical of casework in terms of difficulty. See Appendix K7 for details.

4. Conclusions

This study was conducted to evaluate the accuracy and reproducibility of decisions made by practicing LPEs when comparing latent fingerprints to exemplars that were acquired by searches of the FBI NGI system — as well as to evaluate whether the accuracy and reproducibility of LPE decisions changed in the 13 years since the original LatentBB09 study was conducted, given the potential concern that the higher accuracy of NGI may have an effect on accuracy or reproducibility of LPE decisions. There is a valid concern that highly accurate AFIS with large databases have a serious potential for highly similar nonmated candidates, which then pose a notable risk for FPs: the exemplars returned in response to latent searches using any modern AFIS with a large database will necessarily be much more similar to the latent than exemplars resulting from non-AFIS casework — and this level of similarity can be expected to be greater when the AFIS databases are very large and the algorithms are state of the art. However, we do not find evidence that rates of erroneous IDs have notably increased since LatentBB09: the positive predictive values for the two studies are the same, and although the point estimates for FPR are higher in this study, the confidence intervals overlap for all three (PRES, COMP, and CALLS) approaches, which suggests that the observed rates are consistent between the two studies. The results do suggest that the risk of FPs is higher for participants from agencies that do not provide policies or guidance regarding *IDs* in AFIS vs. non-AFIS casework. The measured FNR (particularly FNR_{CALLS}) is lower in this study than in LatentBB09, which we assume but cannot know may be due to changes in laboratory policies in response to the unexpectedly high FNR in LatentBB09.

More than half of the FPs were made by one participant (who also had an above-average FNR). This participant is a practicing examiner, and neither that person nor their employer (nor we) know who they are. Anonymity was required by the FBI's Institutional Review Board (IRB) for Human Subject Research, including the destruction of identifying information for all participants prior to public release of results. We note that other studies we have conducted have also had outlier participants who are responsible for a disproportionate number of errors: for example, in both the latent print eye-tracking study [28] and the footwear black box study [14], one participant made more than half of the FPs.

These results underscore the fact that point estimates of rates for rare events are fragile: a different mix of participants may affect measured FPRs. Overall rates are an average across all participants: because examiners are not interchangeable, these rates can be disproportionately affected by outlier participants. These rates are estimates that should not be taken as precise measures of the overall population. Studies that rely on volunteers cannot be assumed to be representative of the overall population — and we do not have any way of estimating how many outlier examiners there are in the overall population, particularly given that the fingerprint examiner population is unknown. The study was carefully designed for the samples to be as representative of NGI casework as possible. We do note that the study software differed from the software used by LPEs in casework, which may have had an effect on the results.

This study was conducted to provide information for laboratory managers, forensic practitioners, policy makers, and the legal community regarding the accuracy and reproducibility of decisions made by LPEs on comparisons resulting from searches of a very large, modern AFIS (such as NGI). Some of the results suggest possible areas for improvement:

Most of the participants (including all of the participants who made FPs) were from agencies that do not have policies or guidance on how *ID* decisions should differ when the exemplars come from AFIS vs. non-AFIS casework. This may indicate that the overall latent print discipline is not uniform, with some agencies responding to the risk of highly similar comparisons posed by searches of a large, accurate AFIS, whereas other agencies remain unchanged. These results suggest that risk mitigation strategies for making *IDs* on NGI searches may be working for those agencies that have implemented such strategies, such as recommended in [21][3]. Adoption of such policies by agencies that have not yet done so may be beneficial.

None of the participants who made erroneous *IDs* in the study were aware of having made erroneous *IDs* in casework or on a proficiency test. These results indicate that, in some agencies, there may be a lack of meaningful feedback in LPE practice, suggesting possible areas for improvement in verification of casework, and in proficiency tests.

The participants who had never taken a proficiency test or who work at unaccredited agencies made a disproportionate number of errors in the study. These results may suggest possible areas for improvement regarding requirements for accreditation and proficiency tests.

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CRediT authorship contribution statement

Hicklin R Austin: Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Richetelli Nicole:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis. **Taylor AshLee:** Writing – review & editing, Writing – original draft, Investigation. **Buscaglia JoAnn:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

Declaration of Competing Interest

This work was funded by the Federal Bureau of Investigation (FBI) Laboratory and by the National Institute of Justice via an Interagency Agreement with the FBI Laboratory; AshLee Taylor and JoAnn Buscaglia are employees of the FBI Laboratory. Noblis was funded under a contract award from the FBI Laboratory; R. Austin Hicklin and Nicole Richetelli are employees of Noblis.

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[3] The NIST/NIJ Expert Working Group on Human Factors in Latent Print Analysis Recommendation 3.6 states "When comparing latent prints to exemplars generated through AFIS searches, examiners must recognize the possibility and dangers of incidental similarity. Adjustments such as a higher decision threshold, stricter tolerances for differences in appearance, and explicit feature weighting need to be considered. Modifying quality assurance practices for this scenario also should be considered." [21]

participation of LPEs, and in accordance with the Paperwork Reduction Act under 5 CFR 1320.5(b): OMB Control #1110–0086 (expires 2/28/2026).

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.forsciint.2025.112457.

Data availability

All response data is available in the main text or supplementary materials; because participants and laboratories were assured of anonymity as part of IRB approval, results by participant are summarized or deidentified to prevent reidentification. Fingerprint images obtained through NGI searches that were used in image pairs are not releasable due to privacy and protection of personally identifiable information (PII).

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